Markov Chains

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Contents

Markov chains

Definition

A Markov chain $X = \{X_n \colon n \geqslant 0\}$ is a discrete-time stochastic process with the following properties:

- Each X_n is a discrete random variable that takes values in a finite or countable set S (the state space).
- Markov property: Conditional on the present, the future is independent of the past:

  For all $n \geqslant 0$, and for all $i_0, i_1, \ldots, i_{n-1}, i, j \in S$,

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_1 = i_1, X_0 = i_0)$$
$$= \mathbb{P}(X_{n+1} = j \mid X_n = i) = p_{ij}$$

Transition probabilities

The probability

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i)$$

is referred to as the transition probability from state i to state j.

- We only consider homogeneous chains, where the transition probabilities p_{ij} are constant over time (do not depend on n).
- We have

$$\sum_{j \in S} p_{ij} = \sum_{j \in S} \mathbb{P}(X_{n+1} = j \mid X_n = i) = 1$$

Chapman-Kolmogorov equations

Let $r_{ij}(n) = \mathbb{P}(X_n = j \,|\, X_0 = i)$ be the probability that the chain is in state j at time n, given that the initial state is i.

Theorem (Chapman-Kolmogorov equations)

The following property holds:

$$r_{ij}(n+1) = \sum_{k \in S} r_{ik}(n) r_{kj}(1)$$

More generally,

$$r_{ij}(n+m) = \sum_{k \in S} r_{ik}(n) r_{kj}(m)$$

Chapman-Kolmogorov equations

Proof.

By conditioning on the state of the chain at time n and using the Markov property, we have:

$$
\begin{aligned}
r_{ij}(n+1) &= \mathbb{P}(X_{n+1} = j \mid X_0 = i) \\
&= \sum_{k \in S} \mathbb{P}(X_{n+1} = j \mid X_n = k, X_0 = i)\, \mathbb{P}(X_n = k \mid X_0 = i) \\
&= \sum_{k \in S} \mathbb{P}(X_{n+1} = j \mid X_n = k)\, \mathbb{P}(X_n = k \mid X_0 = i) \\
&= \sum_{k \in S} p_{kj}\, \mathbb{P}(X_n = k \mid X_0 = i) = \sum_{k \in S} r_{ik}(n) r_{kj}(1)
\end{aligned}
$$

Chapman-Kolmogorov equations

If we know the initial probability distribution,

$$(\alpha_i = \mathbb{P}(X_0 = i) \colon i \in S),$$

then the probability that the chain is in state j at time n is

$$\begin{aligned}\mathbb{P}(X_n = j) &= \sum_{i \in S} \mathbb{P}(X_n = j \mid X_0 = i)\, \mathbb{P}(X_0 = i) \\ &= \sum_{i \in S} \alpha_i r_{ij}(n)\end{aligned}$$

Transition probability matrix

If the state space is finite, $|S| = r$, we can consider a square $r \times r$ matrix $\boldsymbol{P} = (p_{ij})$.

- *We refer to $\boldsymbol{P}$ as the transition probability matrix.*
- *We say that $\boldsymbol{P}$ is a stochastic matrix, because for each row i, we have $\sum_{j=1}^{r} p_{ij} = 1$.*

More generally, if $\boldsymbol{P}(n) = (r_{ij}(n))$, then (Chapman-Kolmogorov equations):

$$\boldsymbol{P}(n+1) = \boldsymbol{P}(n) \cdot \boldsymbol{P}, \quad n \geqslant 1$$

Therefore, since $\boldsymbol{P}(1) = \boldsymbol{P}$,

$$\boldsymbol{P}(n) = \boldsymbol{P}^n, \quad n \geqslant 1$$

Transition probability matrix

Let $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_r)$ be the initial probability distribution vector, where

$$\alpha_i = \mathbb{P}(X_0 = i), \quad 1 \leqslant i \leqslant r$$

and let $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_r)$ be the probability distribution vector at time n,

$$\beta_i = \mathbb{P}(X_n = i), \quad 1 \leqslant i \leqslant r$$

Since $\mathbb{P}(X_n = j) = \sum_{i=1}^{r} \mathbb{P}(X_n = j \,|\, X_0 = i)\, \mathbb{P}(X_0 = i)$, we have:

$$\boldsymbol{\beta} = \boldsymbol{\alpha} \boldsymbol{P}^n$$

- In particular, if $\mathbb{P}(X_0 = i) = 1$ (the chain is started in state i), then $\beta = \boldsymbol{\alpha} \boldsymbol{P}^n = (0, \ldots, 0, 1, 0, \ldots, 0)\boldsymbol{P}^n$ gives the i-th row of $\boldsymbol{P}^n$.

Transition probability matrix

Example:

Consider a finite Markov chain with state space $S = \{1, 2, 3\}$, and transition probability matrix

$$\boldsymbol{P} = \begin{pmatrix} 0.2 & 0.7 & 0.1 \\ 0.1 & 0.8 & 0.1 \\ 0.4 & 0.4 & 0.2 \end{pmatrix}$$

Transition probability matrix

Computing $\boldsymbol{P}^2$ we get

$$\boldsymbol{P}^2 = \begin{pmatrix} 0.15 & 0.74 & 0.11 \\ 0.14 & 0.75 & 0.11 \\ 0.20 & 0.68 & 0.12 \end{pmatrix}$$

Thus, for instance,

- $r_{11}(2) = \mathbb{P}(X_2 = 1 \mid X_0 = 1) = 0.15$
- $r_{23}(2) = \mathbb{P}(X_2 = 3 \mid X_0 = 2) = 0.11$
- $r_{33}(2) = \mathbb{P}(X_2 = 3 \mid X_0 = 3) = 0.12$

Transition probability matrix

- If, for example,

$$\mathbb{P}(X_0 = 1) = 0.8,\ \mathbb{P}(X_0 = 2) = 0.1,\ \mathbb{P}(X_0 = 3) = 0.1$$

then the probability distribution of X_2 is given by

$$(0.8\ 0.1\ 0.1)\begin{pmatrix} 0.15 & 0.74 & 0.11 \\ 0.14 & 0.75 & 0.11 \\ 0.20 & 0.68 & 0.12 \end{pmatrix} = (0.154\ 0.735\ 0.111)$$

Transition probability matrix

If we compute $\boldsymbol{P}^n$, we get for $n \geqslant 7$ (rounding to five decimal digits):

$$\boldsymbol{P}^n = \begin{pmatrix} 0.14815 & 0.74074 & 0.11111 \\ 0.14815 & 0.74074 & 0.11111 \\ 0.14815 & 0.74074 & 0.11111 \end{pmatrix}$$

- The probability distribution of X_n seems to converge to the vector

  $$(0.14815 \quad 0.74074 \quad 0.11111)$$

  independently of the initial state of the chain.

Stopping times

A nonnegative integer-valued random variable T is called a stopping time for the Markov chain X if the knowledge of the values of $X_0, X_1, \ldots, X_n$ determines the occurrence or non-occurrence of the event $\{T = n\}$.

For example, the time

$$T_i = \min\{n \geqslant 1 \colon X_n = i\}$$

of the first visit (return) to state i is a stopping time because

$$\{T_i = n\} = \{X_1 \neq i, \ldots, X_{n-1} \neq i, X_n = i\}$$

Strong Markov property

Theorem

Let T be a stopping time for the Markov chain X. Given the events

$$\{T = n\}, \quad \{X_T = i\},$$

any other information about X_0, X_1, …, X_T is irrelevant for predicting the future.

Moreover,

$$\{X_T, X_{T+1}, X_{T+2}, \ldots\}$$

behaves like the Markov chain X with initial state i.

Recurrent and transient states

Example:

Consider a finite chain with state space $S = \{1, 2, 3\}$ and transition probability matrix

$$\boldsymbol{P} = \begin{pmatrix} 0.2 & 0.7 & 0.1 \\ 0.1 & 0.8 & 0.1 \\ 0.4 & 0.4 & 0.2 \end{pmatrix}$$

- Let us consider the time T_1 of the first visit to 1.
- Looking at the first column of $\boldsymbol{P}$, we deduce that

$$\mathbb{P}(T_1 = 1 \mid X_0 = j) \geqslant 0.1 \quad \text{for all } j \in S$$

Recurrent and transient states

Therefore, for all $j \in S$,

$$\mathbb{P}(T_1 > 1 \mid X_0 = j) = 1 - \mathbb{P}(T_1 = 1 \mid X_0 = j) \leqslant 0.9$$

and the Markov property implies:

$$\mathbb{P}(T_1 > 2 \mid X_0 = 1) \leqslant (0.9)^2$$
$$\cdots\cdots\cdots\cdots\cdots$$
$$\mathbb{P}(T_1 > n \mid X_0 = 1) \leqslant (0.9)^n$$

Hence,

$$\mathbb{P}(T_1 > n \mid X_0 = 1) \longrightarrow 0 \quad \text{as } n \longrightarrow \infty$$

Recurrent and transient states

If we denote the event $\{ T_1 > n \,|\, X_0 = 1\}$ by A_n, we have that

$$A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n \supseteq A_{n+1} \supseteq \cdots$$

Thus, the limit event:

$$\begin{aligned} \lim_{n\to\infty} A_n &\equiv \bigcap_{n\geqslant 1} A_n \\ &= \bigcap_{n\geqslant 1} \{ T_1 > n \,|\, X_0 = 1\} = \{ T_1 = \infty \,|\, X_0 = 1\} \end{aligned}$$

is the event "conditional on $\{X_0 = i\}$, the chain does not return to state i ".

Recurrent and transient states

Therefore, in our example,

$$\mathbb{P}(T_1 = \infty \,|\, X_0 = 1) = \mathbb{P}(\lim_{n\to\infty} A_n) = \lim_{n\to\infty} \mathbb{P}(A_n) = 0,$$

or, equivalently,

$$\mathbb{P}(T_1 < \infty \,|\, X_0 = 1) = 1$$

- If the chain starts in state 1, then the eventual return to this state will occur with probability 1.
- By the strong Markov property, the chain will return to state 1 infinitely many times.

Recurrent and transient states

Definition

Let $T_i = \min\{n \geqslant 1 \colon X_n = i\}$ be the time of the first visit to i (being there at time 0 doesn't count), and let

$$f_i = \mathbb{P}(T_i < \infty \mid X_0 = i)$$

be the probability of an eventual return to state i, given that $\{X_0 = i\}$.

The state i is called recurrent if $f_i = 1$. Otherwise, if $f_i < 1$, it is called transient.

- If the Markov chain starts in a recurrent state, this state will be (almost surely) visited infinitely many times (the state continually recurs).

Recurrent and transient states

Example:

Let $S = \{1, 2, 3, 4\}$ and

$$\boldsymbol{P} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0.3 & 0.4 & 0.3 & 0 \\ 0 & 0.3 & 0.4 & 0.3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Since $p_{11} = 1$, we have that $\mathbb{P}(T_2 = \infty \mid X_0 = 2) \geqslant p_{21} = 0.3$

That is,

$$f_2 = \mathbb{P}(T_2 < \infty \mid X_0 = 2) < 1$$

and state 2 is transient.

Recurrent and transient states

Suppose that i is transient and let N_i be the number of visits to i after $n = 0$. Therefore,

$$\mathbb{P}(N_i = k \mid X_0 = i) = f_i^k(1 - f_i), \quad k \geqslant 0$$

(This is a geometric probability distribution starting at $k = 0$.)

Notice that

$$\mathbb{E}(N_i \mid X_0 = i) = \sum_{k=0}^{\infty} k \, f_i^k(1 - f_i) = \frac{f_i}{1 - f_i} < \infty$$

Recurrent and transient states

Let $\{N_i < \infty\} = \bigcup_{k \geqslant 0}\{N_i = k\}$ be the event "the chain visits i only a finite number of times".

Then

$$\begin{aligned}&\mathbb{P}(N_i < \infty \mid X_0 = i)\\ &\quad = \sum_{k=0}^{\infty} \mathbb{P}(N_i = k \mid X_0 = i) = \sum_{n=0}^{\infty} f_i^k(1 - f_i) = 1\end{aligned}$$

- A transient state will be visited only a finite number of times (almost surely).
- In a finite Markov chain it is impossible that all the states are transient. There is at least one recurrent state.

Recurrent and transient states

Theorem

State i is recurrent if and only if

$$\sum_{n=0}^{\infty} r_{ii}(n) = \infty$$

Equivalently, state i is transient if and only if

$$\sum_{n=0}^{\infty} r_{ii}(n) < \infty$$

Recurrent and transient states

Proof:

Let I_n be the indicator random variable of the event $\{X_n = i\}$.

Thus, $\sum_{n=0}^{\infty} I_n$ is the number of times the chain is in state i.

Hence

$$\mathbb{E}\left(\sum_{n=0}^{\infty} I_n \,\middle|\, X_0 = i\right) = \sum_{n=0}^{\infty} \mathbb{E}\left(I_n \mid X_0 = i\right)$$

$$= \sum_{n=0}^{\infty} \mathbb{P}\left(X_n = i \mid X_0 = i\right) = \sum_{n=0}^{\infty} r_{ii}(n),$$

where $r_{ii}(0) = 1$.

The proof is complete by recalling that $\mathbb{E}\left(\sum_{n=0}^{\infty} I_n \mid X_0 = i\right)$ is finite if and only if i is transient.

Recurrent and transient states

If i is a transient state, then $\sum_{n=0}^{\infty} r_{ii}(n) < \infty$, and this condition implies:

$$r_{ii}(n) \to 0 \quad \text{as } n \to \infty$$

More generally,

Theorem

If j is transient and $i \in S$, then $\sum_{n=0}^{\infty} r_{ij}(n) < \infty$. Therefore

$$\lim_{n\to\infty} r_{ij}(n) = 0.$$

Intercommunication

A Markov chain G can be represented by a directed graph G_X in which nodes correspond to states and there is a directed edge from node i to node j if and only if $p_{ij} > 0$.

Definition

State j is reachable from state i if in G_X there is a directed path from i to j.

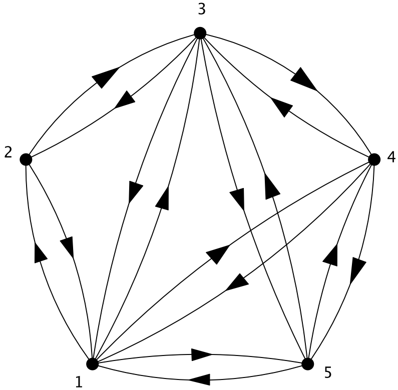
Equivalently,

$$r_{ij}(n) = \mathbb{P}(X_n = j \mid X_0 = i) > 0$$

for some $n \geqslant 0$.

We denote by $A(i)$ the set of states that are reachable from i.

Intercommunication



$$\boldsymbol{P} = \begin{pmatrix} 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 \end{pmatrix}$$

Intercommunication

Definition

The states i and j intercommunicate if $i \in A(j)$ and $j \in A(i)$, in which case we write $i \leftrightarrow j$.

Theorem

If i is a recurrent state and $i \leftrightarrow j$, then j is recurrent. Thus, if i is transient and $i \leftrightarrow j$, then j is transient.

- The relation $\leftrightarrow$ is an equivalence relation. The state space S is partitioned by $\leftrightarrow$ into equivalence classes.
- Within each class all states are recurrent or all are transient.

Intercommunication

Proof:

Since i is recurrent, $\sum_{n=0}^{\infty} r_{ii}(n) = \infty$.

The condition $i \leftrightarrow j$ implies the existence of integers k and m such that $r_{ij}(k) > 0$ and $r_{ji}(m) > 0$. Therefore

$$r_{jj}(m+n+k) \geqslant r_{ji}(m) r_{ii}(n) r_{ij}(k) \quad \text{for all } n \geqslant 0$$

Then,

$$\sum_{n=0}^{\infty} r_{jj}(m+n+k) \geqslant r_{ji}(m) r_{ij}(k) \sum_{n=0}^{\infty} r_{ii}(n) = \infty$$

which implies $\sum_{n=0}^{\infty} r_{jj}(n) = \infty$.

Intercommunication

Definition

A chain is irreducible if $i \leftrightarrow j$ for all $i, j \in S$. (The chain consists of a single class of states that intercommunicate each other.)

Taking into account that a finite chain has at least a recurrent state we have that:

Theorem

If the chain is finite and irreducible, then all states are recurrent.

Mean recurrence time

Definition

The mean recurrence time of the state i is defined as

$$\mu_i = \mathbb{E}\left(T_i \,|\, X_0 = i\right).$$

- Notice that if i is transient, then $\mu_i = \infty$.
- Let i be recurrent. When $\mu_i < \infty$ the state is called non-null or positive. If $\mu_i = \infty$, then i is called null.
- If i is a positive state, then all states of its equivalence class are also positive.

Finite chains: topological sense of recurrence

In a finite chain, the concept of recurrence can be inferred from the topology of the directed graph that represents the chain.

Theorem

A state i of a finite chain is recurrent if and only if, for all $j \in S$,

$$j \in A(i) \Longrightarrow i \in A(j).$$

Finite chains: topological sense of recurrence

- In a finite chain, i being transient means that i is not reachable from some state j which is reachable from i.
- Hence,

$$\mathbb{P}(T_i = \infty \mid X_0 = i) > 0$$

  and i will be visited only a finite number of times.
- It is important to notice that for a chain with an infinite number of states, the condition $i \in A(j)$ for all $j \in A(i)$ does not imply necessarily that $f_i = \mathbb{P}(T_i < \infty \mid X_0 = i) = 1$.

Finite chains: topological sense of recurrence

Example:

Let

$$\boldsymbol{P} = \begin{pmatrix} 0.4 & 0.6 & 0 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.7 & 0.2 \end{pmatrix}$$

State 3 can not be reached from 1 in one step, but we can reach 3 from any state in two steps:

$$\boldsymbol{P}^2 = \begin{pmatrix} 0.28 & 0.54 & 0.18 \\ 0.21 & 0.58 & 0.21 \\ 0.20 & 0.55 & 0.25 \end{pmatrix}$$

Finite chains: topological sense of recurrence

Looking at the third column of $\boldsymbol{P}^2$, we notice that

$$\mathbb{P}(T_3 \leqslant 2 \,|\, X_0 = j) \geqslant 0.18 \quad \text{for all } j \in S$$

Hence,

$$\mathbb{P}(T_3 > 2n \,|\, X_0 = 3) \leqslant (0.82)^n \longrightarrow 0$$

and so

$$\mathbb{P}(T_3 = \infty \,|\, X_0 = 3) = 0$$

Therefore, state 3 is recurrent because

$$f_3 = \mathbb{P}(T_3 < \infty \,|\, X_0 = 3) = 1 - \mathbb{P}(T_3 = \infty \,|\, X_0 = 3) = 1$$

Finite and irreducible chains

Theorem

If the chain is finite and irreducible, then all states are positive.

Proof:

Let $i \in S$ be any state, and let $\beta_m = \mathbb{P}(T_i > m \mid X_0 = i)$. Thus,

$$1 = \beta_0 \geqslant \beta_1 \geqslant \beta_2 \geqslant \ldots \geqslant \beta_m \geqslant \ldots$$

Moreover, since i is reachable from any $j \in S$, we can find an integer k and a number $\alpha \in (0, 1)$, such that

$$\mathbb{P}(T_i \leqslant k \mid X_0 = j) \geqslant \alpha > 0, \quad \text{for all } j \in S$$

Finite and irreducible chains

Therefore, $\mathbb{P}(T_i > k \mid X_0 = j) \leqslant 1 - \alpha < 1$, and, by the Markov property,

$$\beta_{nk} = \mathbb{P}(T_i > nk \mid X_0 = i) \leqslant (1-\alpha)^n, \quad n \geqslant 1$$

Thus,

$$\begin{aligned} \mu_i = \mathbb{E}(T_i \mid X_0 = i) &= \sum_{m \geqslant 0} \mathbb{P}(T_i > m \mid X_0 = i) \\ &= \sum_{m \geqslant 0} \beta_m \leqslant \sum_{n \geqslant 0} k\, \beta_{nk} \leqslant k \sum_{n \geqslant 0} (1-\alpha)^n < \infty. \end{aligned}$$

This result proves that state i is positive.

Finite chains: structure

(Structure)

The following properties hold for a finite Markov chain:

- *The chain can be decomposed into one or more recurrent classes, as well as possibly some transient states.*
- *A recurrent state is reachable from all states within its own class, but not from recurrent states in other classes.*
- *No transient state can be reached from any recurrent state.*
- *From any given transient state, at least one recurrent state is reachable, and possibly more.*

Finite chains: time evolution

(Time evolution)

- *Once the Markov chain enters, or begins in, a class of recurrent states, it remains within that class. Because all states in the class are mutually reachable, each state will be visited infinitely often.*
- *If the initial state is transient, the state trajectory will first pass through transient states before eventually entering and remaining within a single class of recurrent states.*